

Department of Bioengineering
Christian Medical College Vellore
Information sheet

Study Title: HOMER: Home-based robot-assisted upper-limb training for stroke in India

Date:

You are requested to participate in a study to evaluate if robot-assisted home-based arm and hand therapy is better than a standard home-based therapy program prescribed to patients with arm and hand impairments after a stroke. We have developed two robotic devices called PLUTO for assisted training of hand functions and MARS for assisted training of arm functions. We wish to test in this study if training with these devices leads to better recovery than a standard home therapy program that is used for arm and hand therapy. The results of this study can provide evidence for whether we need robots to provide arm and hand therapy at home.

If you take part, what will you have to do?

You will be randomly assigned to either undergo the standard home therapy program for four weeks or you will be provided a robot-based training program at home for four weeks. Your arm and hand movement ability will be assessed three times during the study.

You will undergo the following training and assessment as part of the study.

Training

Standard Home Therapy Program	Robot-Assisted Home Therapy Program
- You will be given training at home for three days by a therapist who will visit you at your home. - You will need to train for four weeks, 2 hours a day, 6 days a week by yourself without the direct supervision of therapy after the first three days. - You will have a weekly Audio/Video call session with the therapist to review your progress.	- You will be given a HANBO and ARMBO robot, which will be set up at your home for four weeks. - A therapist will supervise your therapy with the robots for the first three days at your home. - You will need to train for four weeks, 2 hours a day, 6 days a week by yourself without the direct supervision of therapy after the first three days. You will need to train with the robots for 1.5hrs and 0.5hrs of non-robotic therapy. - You will have a weekly Audio/Video call session with the therapist to review your progress.

Assessments

You will be assessed 3 times as part of this study as described below. Each assessment session will take a maximum of 60 minutes.

- **Assessment 0:** This is a baseline assessment carried out before the start of the training intervention. Here, the ability of your upper limb will be assessed using various standard clinical scales. The amount of movement you possess in your wrist and hand will be evaluated using HANBO and ARMBO.
- **Assessment 1 (Day 30-37):** This assessment will be done immediately after completing the four-week training. In addition to assessing the ability of your upper limb, we will also assess your

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experience with the four-week training either with the robots or the home-therapy program through an in-person interview. This assessment will tell us if your upper limb has improved from the training and your perception of the home therapy program with or without the robot. With your permission, the interview will be audio-recorded to ensure that your responses are captured accurately for analysis. The audio recordings will be kept confidential, securely stored, and accessed only by authorized members of the research team.

- **Assessment 2 (Day 180-187):** This assessment will be done at the end of the 6th month from the day you had your stroke. The ability of your upper limb will be evaluated in this assessment. This will tell us if any gain in upper limb ability you made from the four-week training is maintained as you enter the chronic phase of your stroke.

If you are currently receiving any other therapy except upper limb rehabilitation or exercises, you must continue that in addition to being part of the current study.

Does using the two robots have any side effects?

There are minimal side effects or harm from using these robots such as pain, fatigue, physical discomfort, and eye strain, and no greater than those encountered during the regular therapy session. The robots are safe to use and have all the necessary safety mechanisms to prevent the application of any unsafe forces to your hand. The robots also have emergency stop buttons under your control for completely shutting them down. We have already done short pilot usability studies where no adverse events were reported.

Can you withdraw from this study after it starts?

Your participation in this study is entirely voluntary, and you are free to withdraw your participation at any point. Your withdrawal from the study will not affect your usual treatment at this hospital if you are receiving one.

What will happen if you develop any study-related injuries?

The training you will perform will not be very different from the normal activities that you perform during usual therapy. So, the chances of developing study-related injuries are very low.

Will you have to pay for the study session?

No, you will not be requested to pay for the study. You will be reimbursed for your travel to the hospital for carrying out Assessment- 0, 1, 2.

What happens after studying is over?

The information collected from the study will be used to evaluate the effectiveness of a home-based robot-assisted training protocol with a hand and arm robot. This will allow us to identify the best way to incorporate robot-assisted therapy into home therapy.

Will your personal details be kept confidential?

The results of this study will be published in a medical journal, but you will not be identified by name in any publication or presentation of the results. Videos of your hand/arm during training will be recorded during therapy. These videos will not include your face. These videos might be used as part of publications and presentations of the study results. You can also opt out of this video recording.

If you have any further questions, please ask:

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